

EDUCATION

American University, Washington, DC
Master of Science: Data Science

Loyola University Maryland, Baltimore, MD
Bachelor of Arts: Quantitative Economics & Mathematics

Montgomery College, Rockville, MD
Associate of Science: Mathematics

PROGRAMMING LANGUAGES

- Python
- R
- STATA
- SQL
- HTML
- CSS

ML FRAMEWORK

Supervised Learning: Linear & Logistic Regression, Ridge & Lasso Regression, Decision Trees, Random Forest, Support Vector Machines (SVM)

Unsupervised Learning: Clustering (K-means, Hierarchical), Principal Component Analysis (PCA), Independent Component Analysis (ICA), Canonical Correlation Analysis (CCA), Independent Vector Analysis (IVA)

Deep Learning: Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Generative Adversarial Networks (GANs), Graph Neural Networks (GNNs)

OTHER SKILLS

Data Visualization	Tools	Languages
• Tableau	• Github	• English
• PowerBI	• Excel	• French

PROFESSIONAL SUMMARY

AI/ML Software Engineer experienced in building machine learning solutions, data-driven applications, websites, and mobile applications. Skilled in analyzing complex data to uncover insights, automate processes, and solve practical business problems. Experienced in developing interactive dashboards and user-friendly digital products.

CONFERENCE PRESENTATIONS

Tshiani, D. (2026). Smart Soccer Insights: A Computer Vision Framework for Multi-Class Player Detection and Tracking in Soccer Broadcast Footage. Oral presentation at the Robyn Rafferty Mathias Research Conference.

PROFESSIONAL EXPERIENCE

Artificial Intelligence/Machine Learning Software Engineer

Vastly AI | March 2026 – Present

- Develop and maintain backend services that support biometric identification workflows, including facial image search, computer vision models, fingerprint recognition, and iris recognition.
- Support continuous integration and continuous development processes by validating code changes, resolving test failures, reviewing merge requests, and promoting reliable releases across development environments.
- Monitor application performance and failures in AWS using CloudWatch and Logs Insights, investigate recurring errors, and communicate findings and operational risks during daily scrum meetings.
- Improve service reliability through unit and automated testing, validation of partial batch processing behavior, and strengthening logging and exception-handling workflows.
- Collaborate with software engineers, data scientists, and client stakeholders to translate operational needs into maintainable backend features and technical improvements.

Professor Assistant

American University | Jan 2024 – December 2025

- Provided individual and group mentorship in Natural Language Processing, Computer Vision, and Generative AI concepts and applications.

Quantitative Researcher

American Institute of Research | July 2024 – July 2025

- Reduced manual QA and processing time by 50% through scripted automation, pre-processing, and analysis using R and STATA.
- Applied linear/logistic regression, mixed-effects models, and ANCOVA to evaluate treatment effects for cash-transfer programs and multilingual education interventions. Generated reproducible statistical outputs used directly in partner briefs and policy recommendations for governments and research teams.

Research Assistant

KNG Health Consulting, LLC | April 2023 – July 2024 | North Bethesda, MD

- Conducted statistical and geospatial analyses on Medicare claims (CPT utilization, HOPD reliance), identifying access and cost patterns that informed client recommendations for optimizing healthcare delivery.

FEATURED PROJECTS

Topic Modeling for Democratic Erosion Classification

Tech Stack & Methods: Python, NLTK, Hugging Face

Soccer Recruitment Dashboard with Salary Prediction

Tech stack: R, R-Shiny, tidyverse, caret, ggplot, APIs, Random Forest, XGBoost, Regression | click here for demo: [Player Performance Lab](#)

Creditworthiness & Risk Assessment Analysis

Tech Stack & Methods: R, tidyverse, caret, Random Forest, SMOTE, F1 Score

Olympic Mixed Curling Power Play Optimization Analysis

Tech stack: Python, GitHub